

Keep AI out-of-the-loop

Recommendation: The State Department's Bureau of Arms Control, Nonproliferation, and Stability should codify the November 2024 Biden–Xi affirmation of human control over the decision to use nuclear weapons. The affirmation is currently declaratory-only, and remains undefined and unverified. It should be negotiated into a USA-China framework of AI in nuclear decisions, which would then ideally be extended to the P5.

Schelling argued that surrendering control to an automated system makes the threat of retaliation more credible. Kavka observed that a rational agent can't sincerely intend to follow through on a retaliation that doesn't prevent any damage and only kills millions. For years we have tried to resolve that paradox through a combination of training, loyalty, and duty. But on at least two occasions (Petrov and Arkhipov), humans have overridden procedures that would have (mistakenly!) produced a retaliatory strike. With the benefit of hindsight we consider those decisions to have been correct, but they nevertheless weaken the credibility of second-strike retaliation that supports MAD as a whole.

So from the perspective of defensive posture, those were reliability failures. A natural replacement would be AI which could autonomously respond, since a machine can follow orders without guilt. This immediately produces a tension between deterrence and corrigibility. Every alignment desideratum (corrigibility, interruptibility, a reliable off-switch, human-in-the-loop, predictability) is *anti*-commitment. The ideal agent is docile at baseline but absolutely inexorable once an adversary crosses the line. Posture designers chasing credibility will spec systems that safety engineers are trying to make impossible. And the desire for corrigibility isn't purely academic: the field's own leaders rank losing control of AI as a risk comparable to nuclear war (Statement on AI Risk, 2023); handing nuclear retaliation to AI combines those risks.

Superhuman AI hacking abilities (such as those found in Mythos) accelerate the nuclear command-and-control (NC3) fear cycle. The USA doesn't have to succeed in compromising China's systems; the mere knowledge that said hacker exists, and that even air-gapped systems can be compromised (e.g. Stuxnet), can be destabilizing on its own. The fact that NC3 shares networks with conventional C2 makes espionage almost indistinguishable from disarming-strike prep, and a command system that believes the channels may be compromised will have a strong incentive to pre-delegate retaliation decisions to machines that can't be recalled.

The Biden-Xi statement is a good starting point to build on. But we need to create definitions, scope, and verification paths soon; otherwise force-modernization decisions on both sides will close the option within a few years. Currently, nobody is negotiating it.